Weekly Assignment 1

Complexity Analysis and Recursion

1 Complexity Analysis of Algorithms (50 pts)

Exercise 1 (10 pts)

Arrange the following growth rates in increasing order:

$$n^3 \qquad \log_2(n) \qquad 3^n \qquad n \qquad 1 \qquad n! \qquad 2^n \qquad n\log_2(n) \qquad n^2$$

Exercise 2 (20 pts)

Determine the time complexity $O(f(n))$ for the following functions:

- $f(n) = n + 2n + 3n + \cdots + (n-1)n + n^2$
- $f(n) = \frac{n}{2}\log_2(n) + \frac{n}{3}\log_3(n) + \frac{n}{4}\log_4(n)$
- $f(n) = \log_2(n) + \log_3(n) + \log_4(n)$
- $f(n) = 500n + 100n^{1.5} + 50n\log_{10}(n)$

Exercise 3 (20 pts)

Consider the following C++ code:

```
void trickyFunction(int n)
{
    for (int i = 1; i < n; i *= 2)
    {
        for (int j = 0; j < i; j++)
        {
            cout << i << " ";
        }
    }
}
```

Analyze the time complexity of the function `trickyFunction(n)` justify your answer by deriving the total number of iterations in the algorithm.

2 Recursion (50 pts)

Exercise 4 (10 pts)

Answer the questions:

1. Define recursion, explain the difference between **direct recursion** and **indirect recursion**. Provide an example illustrating each type.
2. Recursive algorithms often have higher computational complexity compared to iterative ones. Explain why, and discuss how to optimize.

Exercise 5 (40pts)

Write **pseudocode** for a recursive function that lists all possible ways to express an integer n as the sum of integers from 1 to n.

Input:
A single integer n, where $1 \leq n \leq 10^6$.

Output:
A list of all possible decompositions of n into sums of integers from 1 to n.

For example:

Input	**Output**
4	4 3 + 1 2 + 2 2 + 1 + 1 1 + 1 + 1 + 1
5	5 4 + 1 3 + 2 3 + 1 + 1 2 + 2 + 1 2 + 1 + 1 + 1 1 + 1 + 1 + 1 + 1

Weekly Assignment 2

Sort Algorithms

Exercise 1: Sorting Algorithm Characteristics

Answer the questions:

1. Explain the differences between **in-place** sorting and **out-place** sorting.
2. Explain the differences between **comparison-based** and **non-comparison-based** sorting.
3. Define the terms, and then give two example sorting algorithms.
   - Adaptability.
   - Stability.

Exercise 2. Complexity Analysis of Sort Algorithms

Fill in the blank:

Algorithm	**Best Case**	**Worst Case**	**Average Case**	**Memory**	**In-place**	**Comparison**	**Adaptability**	**Stability**
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	Yes	Yes	No	Yes
Selection Sort								
Insertion Sort								
Interchange Sort								
Shaker Sort								
Shell Sort								
Heap Sort								
Quick Sort								
Merge Sort								
Counting Sort								
Radix Sort								

Exercise 3. Choose the correct answer

1. Which sorting algorithm repeatedly swaps adjacent elements if they are in the wrong order?

   A) Bubble sort

   B) Insertion sort

   C) Selection sort

   D) Heap sort

2. Which of the following statements is true about Bubble Sort?

   A) Bubble sort has a worst-case time complexity of $O(n \log n)$

   B) Bubble sort requires $O(n)$ additional memory space

   C) Bubble sort is a stable sorting algorithm

   D) Bubble sort is not an in-place algorithm

3. Which of the following is true about Binary Search?

   A) Binary search can be applied to an array that is not sorted.

   B) Binary search halves the search range with each comparison.

   C) In the worst case, binary search examines every element in the array.

   D) The time complexity of binary search grows linearly with the number of elements.

4. Which sorting algorithm is the most efficient for sorting a list that is already nearly sorted?

   A) Insertion sort

   B) Selection sort

   C) Quick sort

   D) Heap sort

5. Which sorting algorithm would be a good choice for a large list?

   A) Selection Sort

   B) Insertion Sort

   C) Binary Insertion Sort

   D) Quick sort

6. After building a max-heap from the elements of the array `[4, 10, 3, 5, 1]`, which of the following represents the array in heap order?

   A) `[10, 4, 3, 5, 1]`

   B) `[5, 10, 3, 4, 1]`

   C) `[10, 5, 3, 4, 1]`

   D) `[10, 5, 4, 3, 1]`

7. A bio-informatics team needs to sort billions of short DNA sequences (strings) to efficiently find matching sequences. Which sorting algorithm is most efficient?

   A) Merge sort

   B) Quick sort

   C) Counting sort

   D) Radix sort

8. Through experiment, you determine that selection sort performs 5000 comparisons when sorting an array of some size k. If you doubled the size of the array to $2k$, approximately how many comparisons would you expect it to perform?

   A) 5000

   B) 10000

   C) 20000

   D) 40000

   E) The value would depend on the array's content

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Exercise 4. Sorting Algorithm Analysis

Among the sorting algorithms with time complexity $O(n^2)$, which algorithm(s) are the best suited for each of the following situations? Justify your answer.

a) When all elements of the input are identical, which algorithm(s) will take the least time?

b) When swapping is very costly, which algorithm(s) should be preferred to minimize the number of swaps in general?

Exercise 5. Sorting an Array of Numbers

Given the following array of non-negative integers:

```
[100, 77, 49, 1, 29, 51, 7, 15, 90]
```

Perform sorting process on this array using four algorithms: (1) Bubble Sort, (2) Interchange Sort, (3) Insertion Sort, and (4) Selection Sort, by displaying the state of the array after each iteration to illustrate how the algorithm works.

Exercise 6. Sorting an Array of Letters

Given the following array of lowercase letters:

```
[D, S, A, H, C, M, U, S]
```

Perform sorting process on this array using three algorithms: (1) Heap Sort, (2) Merge Sort, and (3) Quick Sort. Show each step of the sorting process to illustrate how the elements are gradually sorted at each stage.

Exercise 7. Sorting an Array of Fruit Names

Given the following array of fruit names (each name is a single lowercase word without spaces):

```
[Banana, Orange, Mango, Peach, Plum, Kiwi, Cherry, Papaya, Fig, Guava]
```

Perform the sorting process on this array using the Radix Sort algorithm by displaying each step of the sorting process, sorting by each letter from the last letter of each word to the first letter.

Weekly Assignment 3

Search Algorithms and Hash Table

Exercise 1. Linear Search

Consider the following code implementing linear search:

```
int linearSearch(int arr[], int size, int target)
{
    int i = 0;
    while (i < size)
    {
        if (arr[i] == target)
            return i;
        i++;
    }
    return -1;
}
```

In the worst-case scenario, how many comparisons does the above code perform.
Modify the code to reduce the number of comparisons by approximately half.

Exercise 2. Binary Search

Given a sorted array with the following 16 elements:

3	7	12	18	21	25	32	40	45	53	60	68	75	82	90	97

If we do binary search for the number **50**, how many elements are checked in the array?
List the elements in the array that are checked during the search.

Exercise 3. Linear Search and Binary Search

Linear search must examine, on average, half of the N elements of an array while binary search needs to only check $log_2(N)$. Thus, binary search is a great deal more efficient than linear search. Why, then, should we still study and use linear search?

Exercise 4. Hash function

Suppose there is a hash table containing several elements, as shown in the figure below:

0		$\rightarrow 0 \rightarrow 30$
1		
2		$\rightarrow 21 \rightarrow 31 \rightarrow 1$
3		
4		$\rightarrow 2$
5		
6		$\rightarrow 33 \rightarrow 63$
7		
8		
9		
10		$\rightarrow 15$
11		
12		$\rightarrow 26 \rightarrow 6$
13		
14		
15		
16		$\rightarrow 38 \rightarrow 8$
17		
18		
19		

Observe the content of the hash table and discover a major problem with the hash table.

- What is the problem with the hash table?
- Describe a hash function that can lead to this problem.
- Suggest another hash function to overcome the above issue.

Exercise 5. Collision resolution strategies

Compare two collision resolution strategies, separate chaining and open addressing.

Exercise 6. Hash Table

Consider the following hash table of size $M = 11$ and a hash function:

$$h(x) = x \mod M$$

Insert the following keys into the hash table in the given order:

$$0, 1, 8, 9, 52, 44, 56, 53, 61, 64$$

using one of the following collision resolution strategies:

1. **Linear Probing**: If a collision occurs, probe sequentially to the next available slot.
2. **Quadratic Probing**: If a collision occurs at position $h(x)$, the next probe attempts positions according to the sequence:

   $$h(x), h(x) + 1^2, h(x) + 2^2, h(x) + 3^2, \ldots \mod M$$

   until an empty slot is found.
3. **Double Hashing**: Use a secondary hash function to resolve collisions:

   $$h'(x) = 7 - (x \mod 7)$$

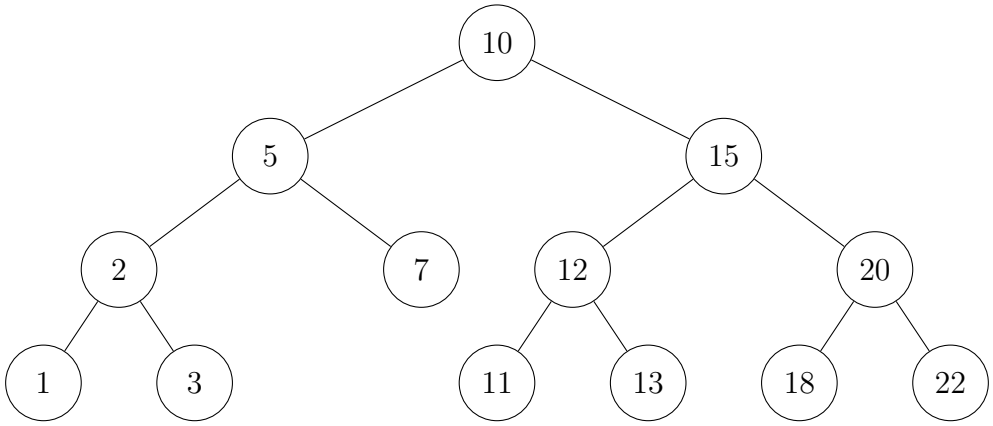
Construct the resulting hash table for each collision resolution strategy and discuss the efficiency of each method.

Weekly Assignment 4

Trees

Binary Tree

Exercise 1 to 3 refer to the following tree:



Exercise 1: Tree Terminologies

Choose the correct answer:

1. How many **nodes** does the given tree have?

   A) 12

   B) 13

   C) 14

   D) 15

2. How many **internal nodes** does the given tree have?

   A) 5

   B) 6

   C) 7

   D) 8

3. How many **external nodes (leaves)** does the given tree have?

   A) 5

   B) 6

   C) 7

   D) 8

4. Which nodes is the **root** of the tree?

   A) 1

   B) 10

   C) 11

   D) 22

5. Which nodes is the **parent** of the node 11?

   A) 7

   B) 10

   C) 12

   D) 13

6. Which of the following lists contains all **children** of node 5?

   A) 2, 7

   B) 1, 2, 3, 7

   C) 1, 2, 3, 5, 7

   D) No correct answer

7. Which pair of nodes are siblings?

   A) (3, 7)

   B) (3, 11)

   C) (5, 15)

   D) (18, 20)

8. Which of the following lists contains all ancestors of node 3?

   A) 2

   B) 1, 2

   C) 2, 5, 10

   D) 2, 5, 7, 10

9. Which of the following lists contains all descendants of node 15?

   A) 12, 20

   B) 11, 12, 13, 18, 20, 22

   C) 11, 12, 13, 15, 18, 20, 22

   D) No correct answer

Exercise 2: Tree Traversals

Traverse the given tree using these strategies:

- Pre-order
- In-order
- Post-order
- Level-order

Which traversal strategy will give us an increasing list of items? Why?

Exercise 3: Types of Binary Tree

Define each of these trees and then provide an example figure.

- Full binary tree
- Complete binary tree
- Perfect binary tree
- Heap

Which type of binary tree is the given tree? Why?

Binary Search Tree

Exercise 4

Construct an BST by sequentially **inserting** each element from left to right in the order they appear in the following list (draw the tree step by step):

42, 17, 89, 3, 76, 54, 25, 61, 8, 97

How many comparisons are needed to **find** the item **54**. Write down the sequence of nodes that might be visited during the search.
Delete the number **61**, **54** and **42** from the constructed tree (draw the tree step by step).

AVL Tree

Exercise 5

Construct an AVL Tree by sequentially **inserting** each element from left to right in the order they appear in the following list (draw the tree step by step):

V, I, C, T, O, R, Y

Delete the letter **I** from the constructed tree (draw the tree step by step).

The end.